

2014 Notes from the Marking Centre – Chemistry

Introduction

This document has been produced for the teachers and candidates of the Stage 6 Chemistry course. It contains comments on candidate responses to the 2014 Higher School Certificate examination, highlighting their strengths in particular parts of the examination and indicating where candidates need to improve.

This document should be read along with:

- the [Chemistry Stage 6 Syllabus \(PDF\)](#)
- the [2014 Higher School Certificate Chemistry examination \(PDF\)](#)
- the [marking guidelines \(PDF\)](#)
- [Advice for students sitting for HSC science examinations](#)
- [Advice for HSC students about examinations](#)
- other support documents developed by the Board of Studies, Teaching and Educational Standards NSW to assist in the teaching and learning of Chemistry in Stage 6.

Section I

Part B

Candidates showed strength in these areas:

- interpreting nuclear equations and identifying methods of producing transuranics (Q.21)
- identifying water contaminants (Q.23)
- knowing a test for an ion (Q.24)
- calculating moles and mass (Q.25 (b))
- naming an ester and describing risks and precautions (Q.26)
- describing Le Chatelier's principle (Q.27)
- using the table of standard potentials to write half equations and calculate E^0 (Q.28).

Candidates need to improve in these areas:

- constructing appropriate scales and drawing lines of best fit for graphs (Q.22)
- understanding the difference between validity, reliability and accuracy of data (Q.22 (b))
- identifying ions using a correct sequence of tests and writing balanced equations (Q.24)
- describing AAS and not confusing it with flame tests and mass spectrometry (Q.27)
- distinguishing between copper atoms (solid) and copper ions (aqueous) (Q.28)
- describing limitations of a model and clearly linking models to real-life situations and experiments (Q.29).

Section II

Question 32 – Industrial Chemistry

Candidates showed strength in these areas:

- balancing equations
- relating the structure of soaps and detergents to their uses and impacts (part (e)).

Candidates need to improve in these areas:

- using correct formulae for species, for example Cl_2 and Cl^-
- specifying quantities, substances and equipment when outlining a practical experience
- addressing the requirements when outlining a procedure, for example focusing on modelling a step in the Solvay process rather than preparation of the reactants, such as brine purification (part (d)).

Question 33 – Shipwrecks, Corrosion and Conservation

Candidates showed strength in these areas:

- identifying controlled variables (part (d) (i))
- identifying factors that affect electrolysis (part (e))
- identifying the artefact as the cathode that undergoes reduction (part (e))
- understanding protection against corrosion.

Candidates need to improve in these areas:

- addressing the requirements when describing limitations, for example focusing on limitations in making observations rather than general limitations of the investigation (part (d) (ii))
- writing half and overall equations (balanced, with states and with reduction potentials).

Question 34 – The Biochemistry of Movement

Candidates showed strength in these areas:

- interpreting a graph (part (c))
- calculating formula mass (part (c) (ii))
- understanding the importance of mitochondria in energy production (part (e)).

Candidates need to improve in these areas:

- identifying the steps in a flow chart (part (a))
- describing two limitations of using models to represent fatty acid molecules (part (d) (ii)).

Question 35 – The Chemistry of Art

Candidates showed strength in these areas:

- understanding the trends in ionisation energy (part (a))
- knowing what pigments were used in early Aboriginal art (part (b) (i))
- understanding electronegativity.

Candidates need to improve in these areas:

- demonstrating knowledge of a first-hand investigation that can be used to study the changes in oxidation state of a transition metal (part (d))
- understanding the difference between absorption and emission (part (e)).

Question 36 – Forensic Chemistry

Candidates showed strength in these areas:

- identifying errors in forensic sampling (part (a))
- identifying an unknown substance using mass spectrum and mass to charge ratio (part (c) (i))
- describing the limitations of a procedure (part (d) (ii)).

Candidates need to improve in these areas:

- drawing amino acids (part (b) (i))
- understanding the principles of DNA evidence collection and analysis.